MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS)
Regular EXAMINATION March 2026

Fourth Semester

Discipline Specific Elective (DSE) – MG4DSEFSQA01 – FOOD SAFETY PRE-REQUISITES (2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Part A
Short Answers
(2 Marks Each)
Answer in one or two sentences. Attempt any 10 questions.

1. Floors should be smooth, impervious, non-absorbent, non-slip, easy to clean and resistant to chemicals with proper drainage.

2. Cloakrooms provide storage for personal belongings and help prevent contamination of food processing areas.

3. Key PRP areas: infrastructure and layout, utilities, cleaning and sanitation, pest control, personnel hygiene, supplier control.

4. Supplier evaluation criteria: regulatory compliance, quality consistency, certification status, delivery performance, traceability.

5. Clutter prevention is the removal of unnecessary items and proper organization of workplace materials to maintain hygiene and efficiency.

6. Separation of non-food items prevents contamination and ensures food safety.

7. Receiving records document source, quantity, quality and condition of incoming materials and support traceability.

8. Stages of sewage treatment: preliminary, primary, secondary, tertiary and sludge treatment.

9. Common pests: rodents, insects, birds and reptiles/other nuisance animals.

10. Protecting food from deliberate contamination safeguards public health and company reputation.

11. Training programs improve employee competence, food safety awareness and compliance.

12. Detergents remove dirt, grease and food residues and facilitate effective sanitation.

Part B – Short Essay Questions (5 Marks Each)
Answer any 6 questions. Each question carries 5 marks.

13. Integration of preventive maintenance with PRPs:
- Prevents equipment breakdown.
- Reduces contamination risks.
- Maintains hygienic design.
- Supports calibration and monitoring.
- Ensures uninterrupted production.
- Requires documentation and records.

14. Importance of equipment calibration:
- Accurate monitoring.
- Correct measurements.
- Regulatory compliance.
- Product consistency.
- Reliable CCP monitoring.
- Reduced recalls.

15. Food-contact materials should be non-toxic, corrosion resistant, smooth, non-porous, durable and easy to clean. Stainless steel is widely used.

16. Uses of compressed air:
- Product conveying.
- Packaging.
- Pneumatic controls.
- Bottle cleaning.

Risks:
- Microbial, oil, moisture and particulate contamination.

Controls:
- Filtration, dryers, oil-free compressors and monitoring.

17. Allergen labelling requirements:
- Declare allergens in ingredient list.
- Use clear identification.
- Mention allergen sources.
- Comply with regulations.

Examples: milk, egg, fish, peanuts, wheat and soy.

18. Environmental impacts of improper waste disposal:
- Soil contamination.
- Water pollution.
- Air pollution.
- Greenhouse gas emissions.
- Pest attraction.
Control through segregation, recycling and scientific disposal.

19. Personal hygiene practices:
- Hand washing.
- Protective clothing.
- Hair restraints.
- No jewellery.
- Cover wounds.
- Medical fitness.
- No eating or smoking in processing areas.

20. Environmental management in pest control:
- Good housekeeping.
- Waste management.
- Elimination of stagnant water.
- Structural maintenance.
- Monitoring and inspections.

Part C – Essay Questions (10 Marks Each)
Answer any 2 questions. Each question carries 10 marks.

21. Hygienic site selection and zoning:
- Away from pollution sources and flooding.
- Adequate water and waste disposal.
- Proper segregation into low, medium and high-risk zones.
- Prevents cross-contamination and supports HACCP.

22. FIFO and FEFO:
FIFO: First In First Out.
FEFO: First Expired First Out.
Importance:
- Reduces spoilage.
- Maintains quality.
- Supports traceability.
- Improves inventory management.

23. Labelling requirements:
- Name of food.
- Ingredient list.
- Nutrition information.
- Veg/non-veg symbol.
- Net quantity.
- Batch number.
- Date marking.
- Manufacturer details.
- FSSAI licence number.
- Allergen declaration.

24. Sanitation chemicals:
- Chlorine compounds.
- Quaternary ammonium compounds.
- Iodophors.
- Peracetic acid.
- Hydrogen peroxide.
- Ozone.

Include applications, advantages and limitations of each.
compliance and records. Prevents misuse, contamination, and accidents.